Student Performance Analyzer Report

By Prachi Bhowal
CSE (24070721027)
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1. Project Overview

The Student Performance Analyzer is a Python and MySQL based mini project developed to manage student academic records through a role-based desktop application. The system supports three roles: Admin, Faculty, and Student. It includes login authentication, student record management, analytics, audit logging, and subject-related views as required in the assignment brief.

2. Design Choices

The project was designed by separating the application into multiple Python modules so that authentication, student operations, analytics, user management, and subject handling are easier to maintain and test. A Tkinter GUI was used for the interface, while MySQL was used as the backend database to store users, students, audit logs, and related academic data. The role-based menu structure was chosen so each user sees only the operations relevant to that role, such as full management access for Admin, limited academic operations for Faculty, and read-only personal access for Students.

Another important design choice was the use of hashed passwords and audit logging. The assignment requires secure login using SHA-256 hashing and recording login or data-changing events in the audit log, so the implementation was planned around those security requirements from the beginning. The project also includes MySQL application users and GRANT-based privilege separation to reflect database-level role access, which aligns with the DBMS-focused objective of the mini project.

3. Features Implemented

- Role-based login system for admin, faculty, and student.
- Student CRUD operations: add, view, search, update, and delete.
- View own record for student users.
- Live dashboard showing total students, average marks, highest marks, and lowest marks.
- Top performers list.
- Marks distribution and department-wise analytics.
- Department leaderboard.

- Add bonus marks by department.
- Export high performers to CSV.
- User management for admin: create user, list users, toggle user status.
- Create MySQL users with role-based privileges.
- Audit log to track important actions.
- Subject-related views for faculty and student marks.

4. Challenges and Solutions

- **Challenge:** Too many buttons in the admin menu made the window look crowded.
  Solution: Added a scrollable sidebar so all options stay accessible even in a small window.
- **Challenge:** Some subject functions were missing, which caused errors during login.
  Solution: Added the missing methods and connected them properly to the subject module.
- **Challenge:** Keeping separate access for admin, faculty, and student.
  Solution: Built the dashboard menu dynamically based on the logged-in role.
- **Challenge:** Packaging the app into an executable.
  Solution: Used PyInstaller to create a standalone .exe file.
- **Challenge:** Protecting passwords and important actions.
  Solution: Used hashed passwords and audit logging.

5. Use of AI

- Used AI for debugging error messages.
- Used AI for understanding and fixing code structure issues.
- Used AI to help explain database concepts clearly.
- Used AI to draft report content and improve wording.
- Final code and testing were still checked manually before submission.

6. Conclusion

The project meets the main goal of building a Student Performance Analyzer using Python and MySQL with role-based access control, database operations, analytics, and audit tracking. It also demonstrates both application-level user management and database-level privilege concepts, making it a suitable DBMS mini project submission.